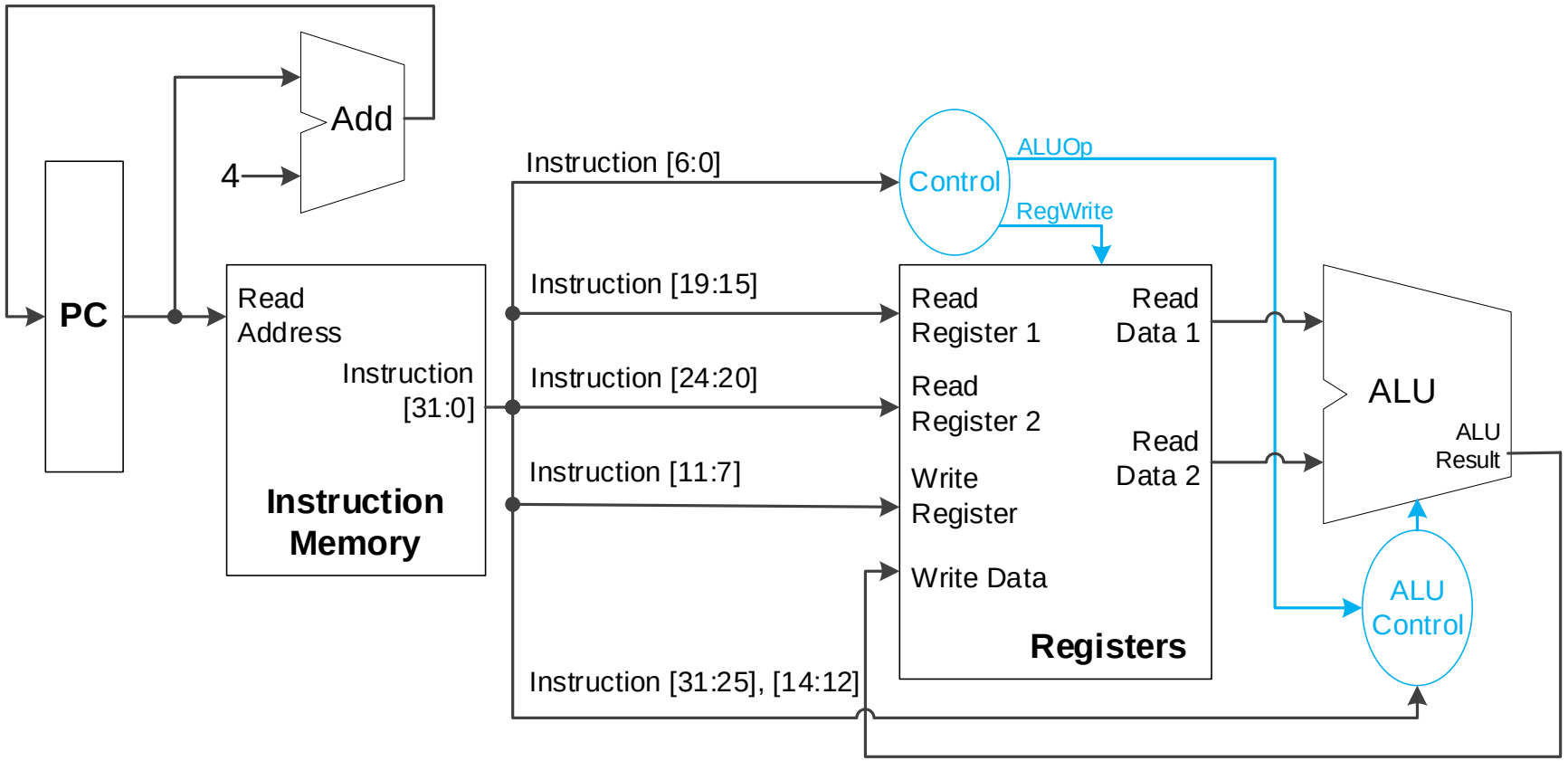
Introduction to Computer Architecture
Chapter 4 - 2

The Processor

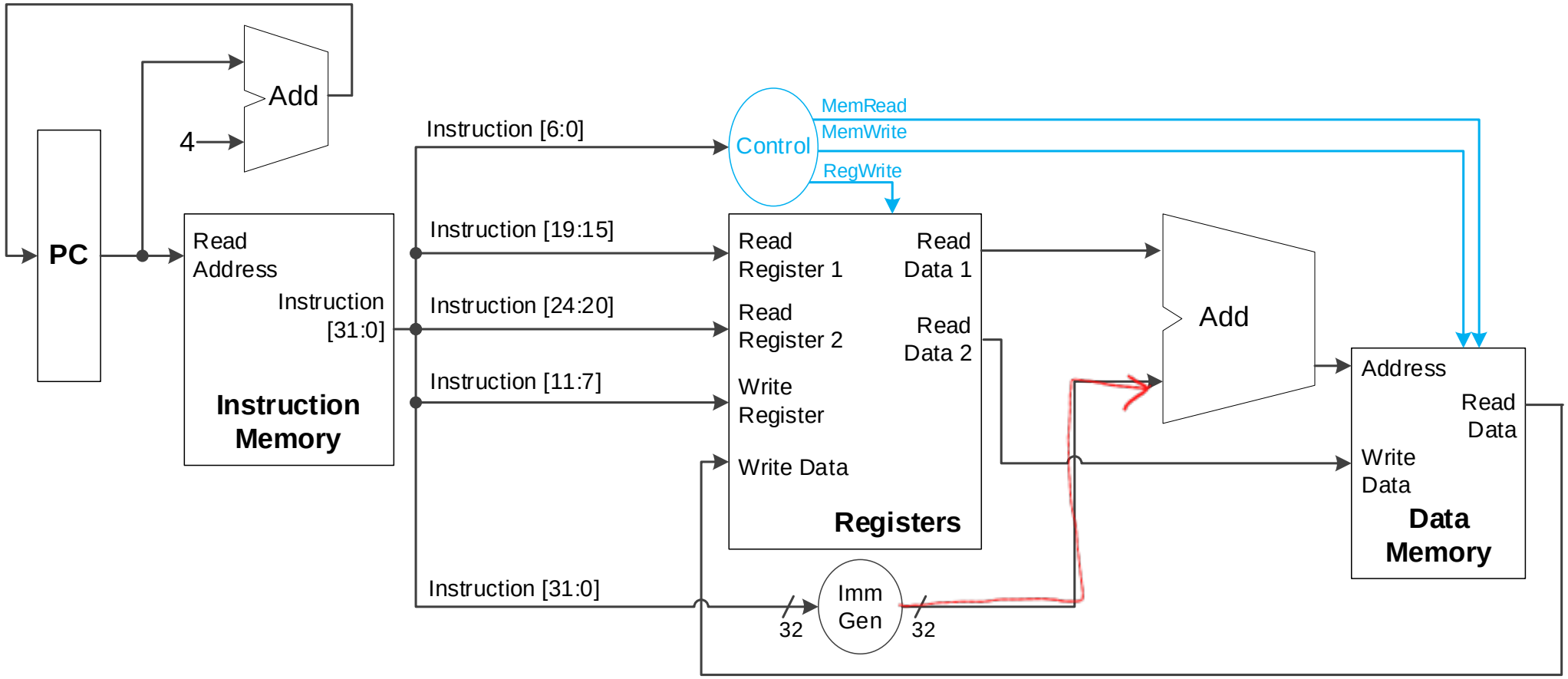
Hyungmin Cho

Department of Computer Science and Engineering
Sungkyunkwan University

CPU for R-type Instructions

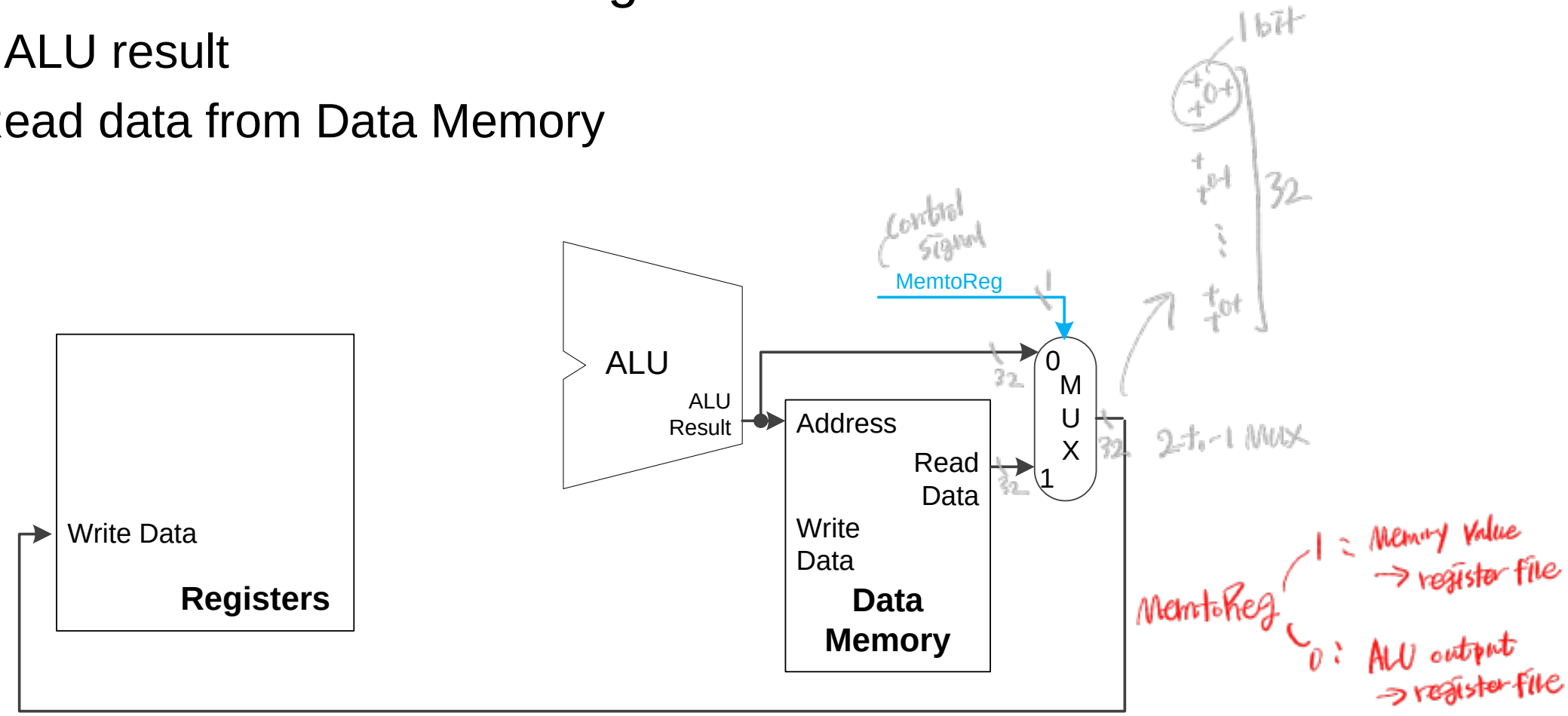


CPU for Load/Store



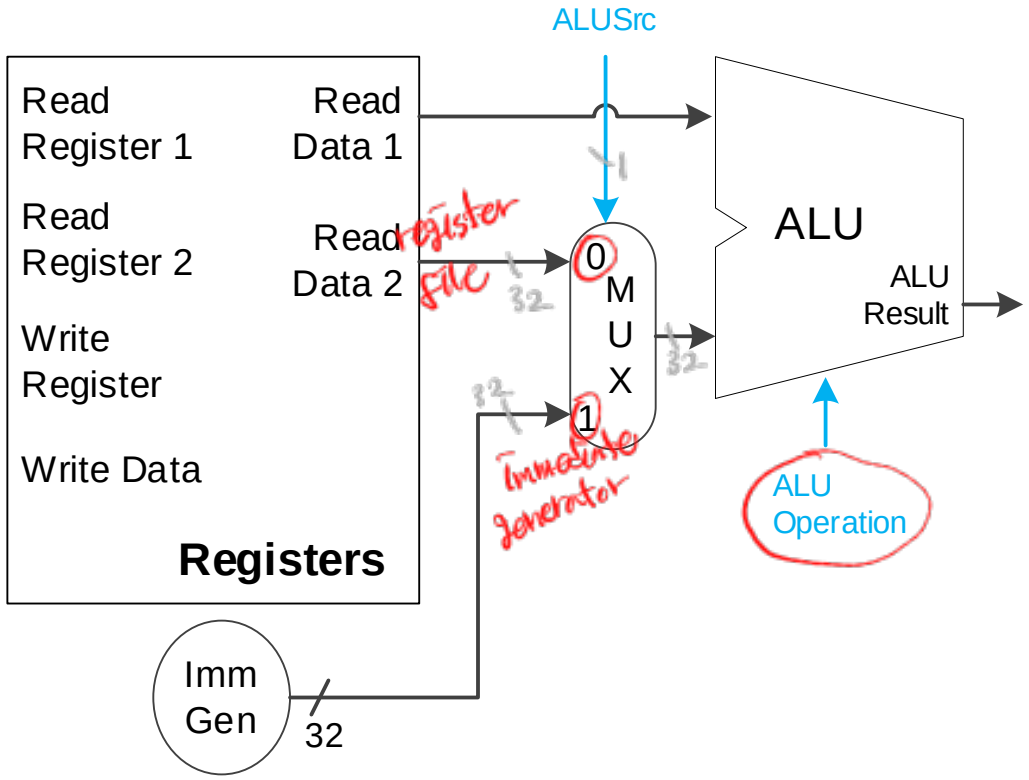
Combining R-type and Load/Store

- What to write in the destination register?
  - R-type: ALU result
  - Load: Read data from Data Memory



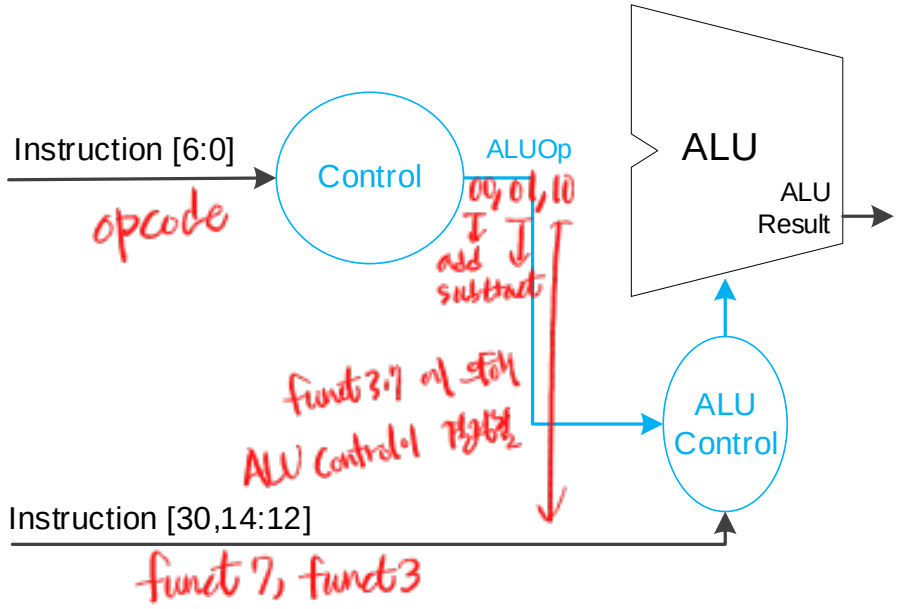
Combining R-type and Load/Store

- One ALU is used for both…
  - R-type Arithmetic/Logic operations
  - Load/Store address calculation

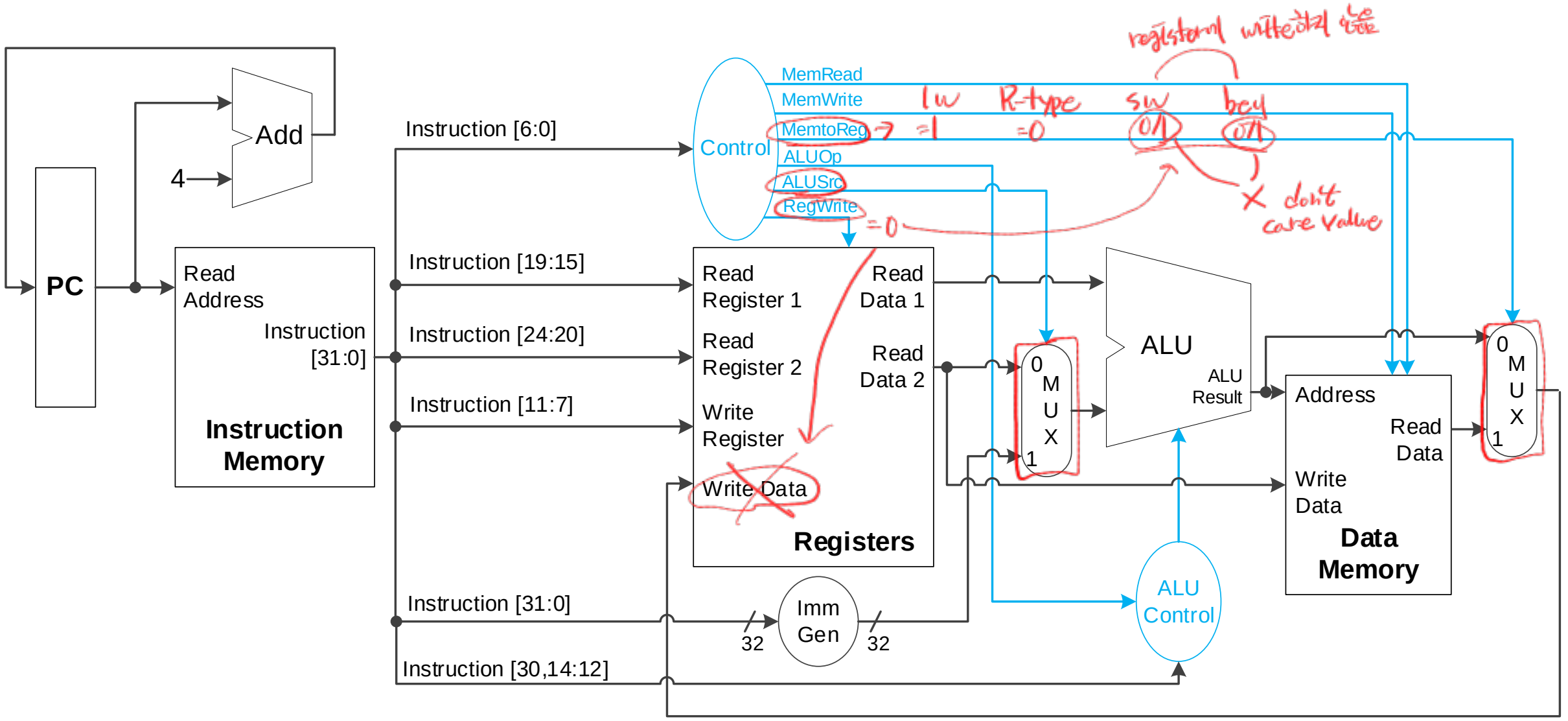


ALU Control Bits

Opcode	ALUOp	Operation	Funct7	Funct3	ALU function	ALU control
lw	00	load word	XXXXXXX	XXX	add	0010
sw	00	store word	XXXXXXX	XXX	add	0010
beq	01	branch if equal	XXXXXXX	XXX	subtract	0110
R-type	10	add	0000000	000	add	0010
		sub	0100000	000	subtract	0110
		and	0000000	111	AND	0000
		or	0000000	110	OR	0001



Combining R-type and Load/Store



Load Execution Example

Register	Data
x1	0x000000FF
x2	0x00000011
x29	0x00001200

Address	Data
0x1240	0x80
0x1241	0x90
0x1242	0xA0
0x1243	0xB0
0x1244	0xC0
0x1245	0xD0

Address	Data
0x3FF8	0x041EA103

Opcode:0000011_2 lw

PC+4: 0x3FFC

Current PC: 0x3FF8

MemRead:1

MemWrite:0

MemtoReg:1

ALUOp: 00_2 = Add

ALUSrc: 1

RegWrite:1

Read data 1: 0x1200

ALU result: 0x1241

ALU Control: 0010_2 = Add

Read data (Mem): 0xC0B0A090

Read data 2: 0xFF

$rs1:11101_2$

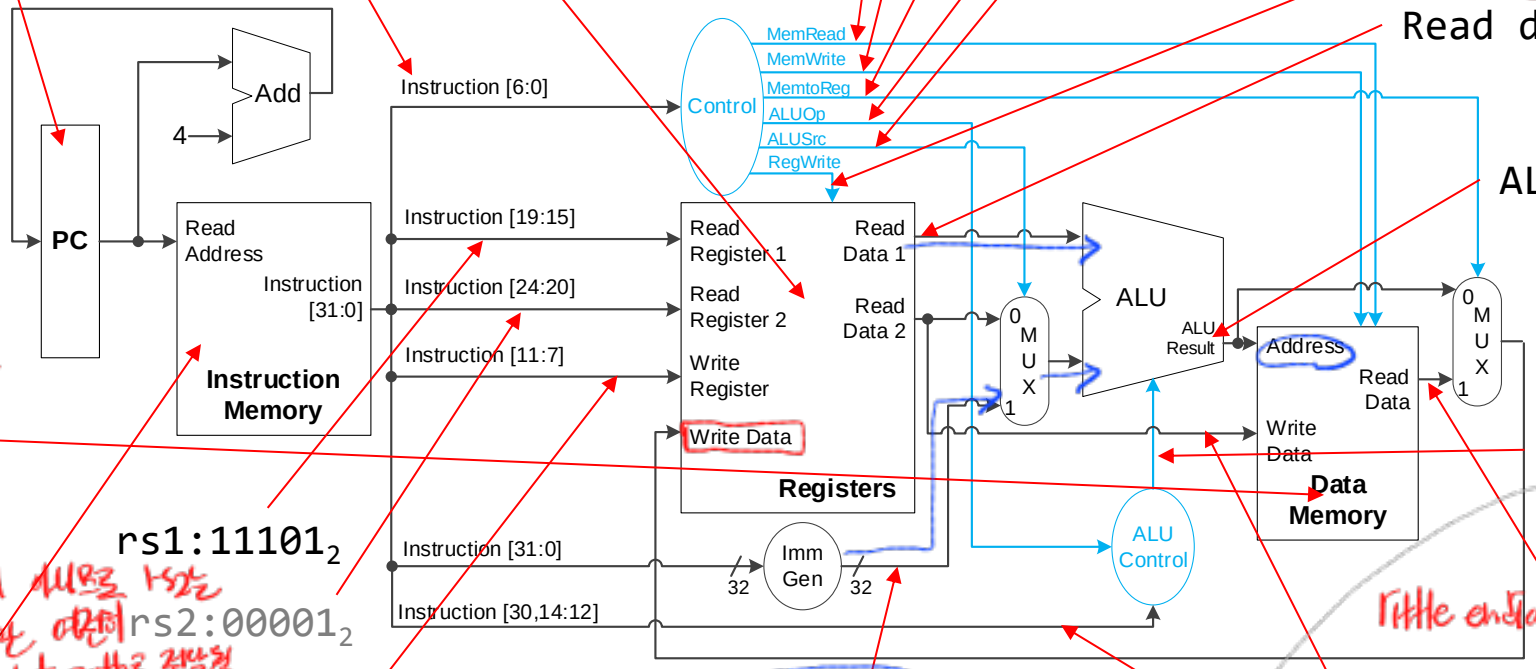
$rs2:00001_2$

$rd:00010_2$

0x00000041

Funct3: 010

Funct7[5]: 0

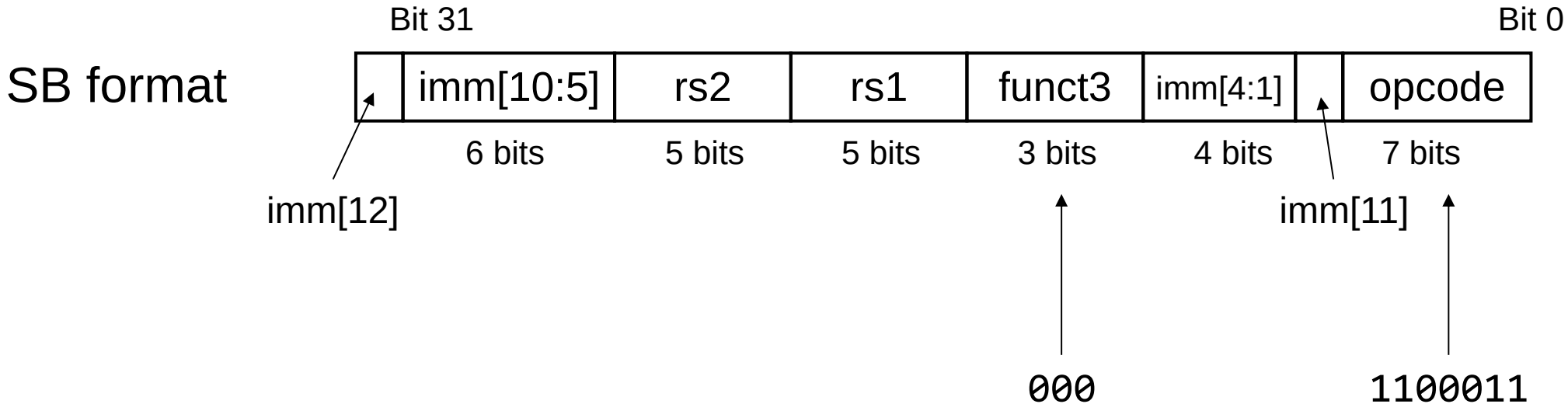


0000 0100 0001 1110 1010 0001 0000 0011 → lw x2, 65(x29)

000001000001 11101 010 00010 0000011 → lw x2, 0x41(x29)

Imm[11:0] rs1=x29 funct3 rd opcode load

Branch Instructions (BEQ)

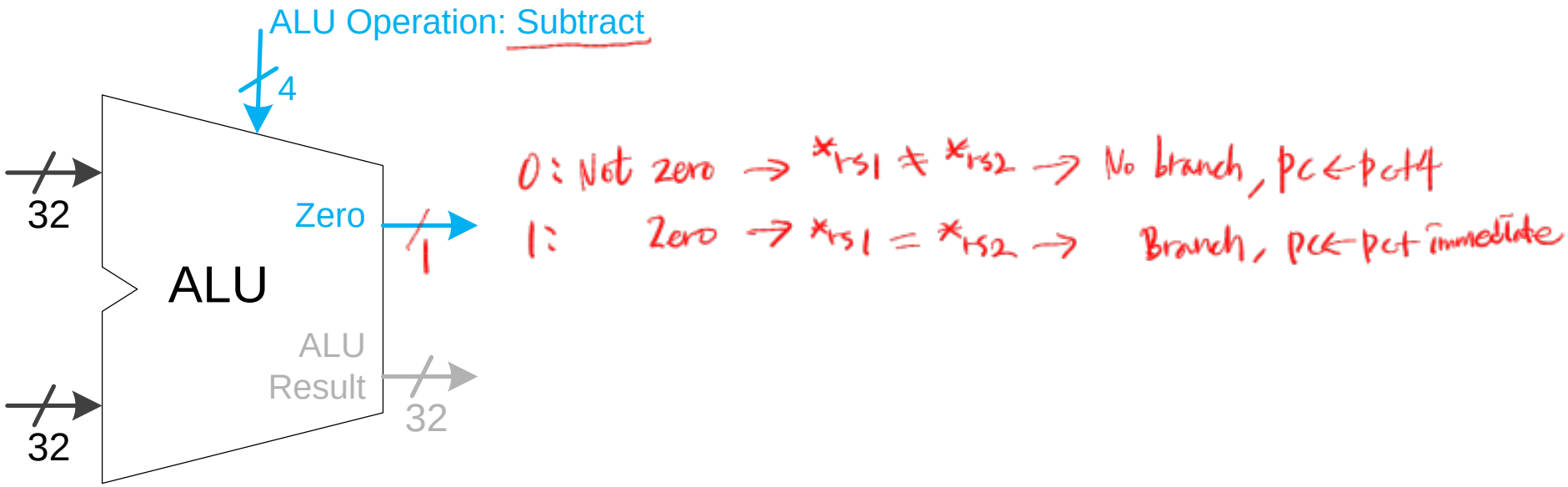


- If `*rs1 == *rs2`
  - New PC = PC + immediate

- Immediate is a 13-bit signed value, with `imm[0]` is assumed to be zero.
  - Need to sign-extend

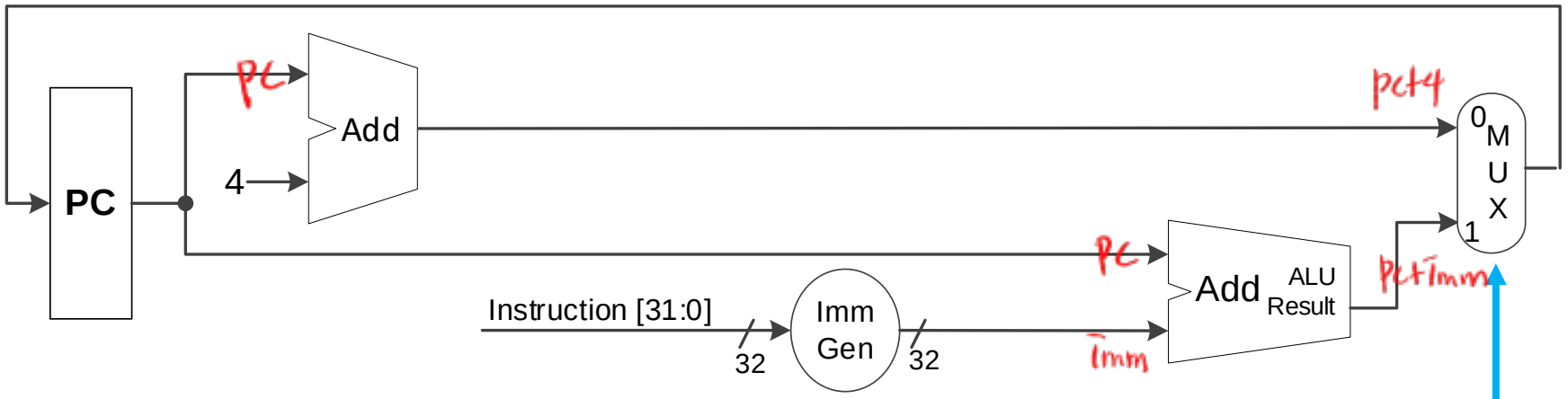
Branch Condition Test

- Subtract and check if the result is zero



- ALU has one extra output – result is equal to zero?

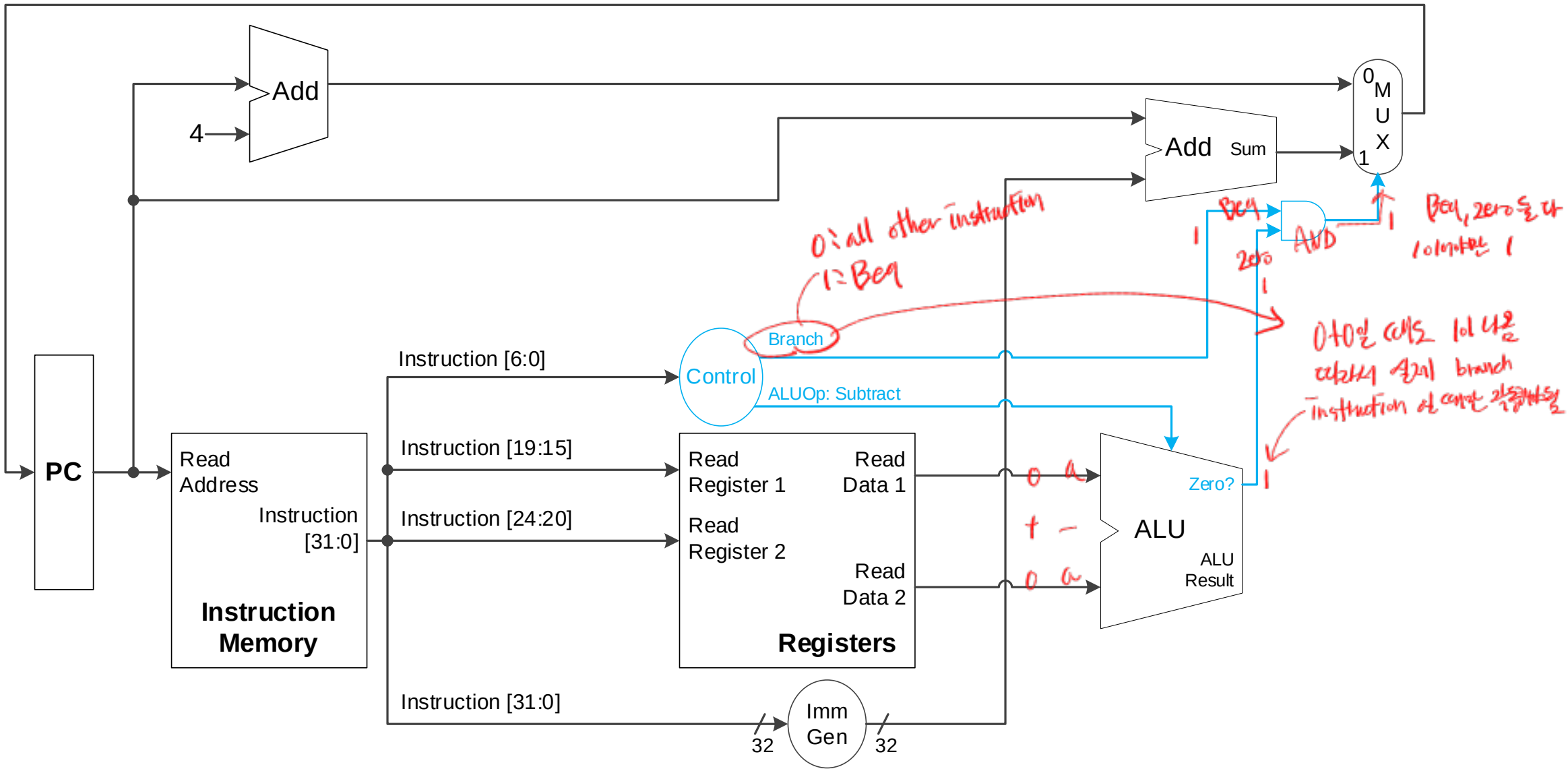
Branch Target (New PC) Calculation



1: zero is set, and the instruction is ‘beq’ branch

0: zero is not set or instruction is something else..

CPU for Branch Instruction



BEQ Execution Example

Register	Data
x8	0x1
x9	0x1

Branch:1

ALUOp:01_2 = subtract

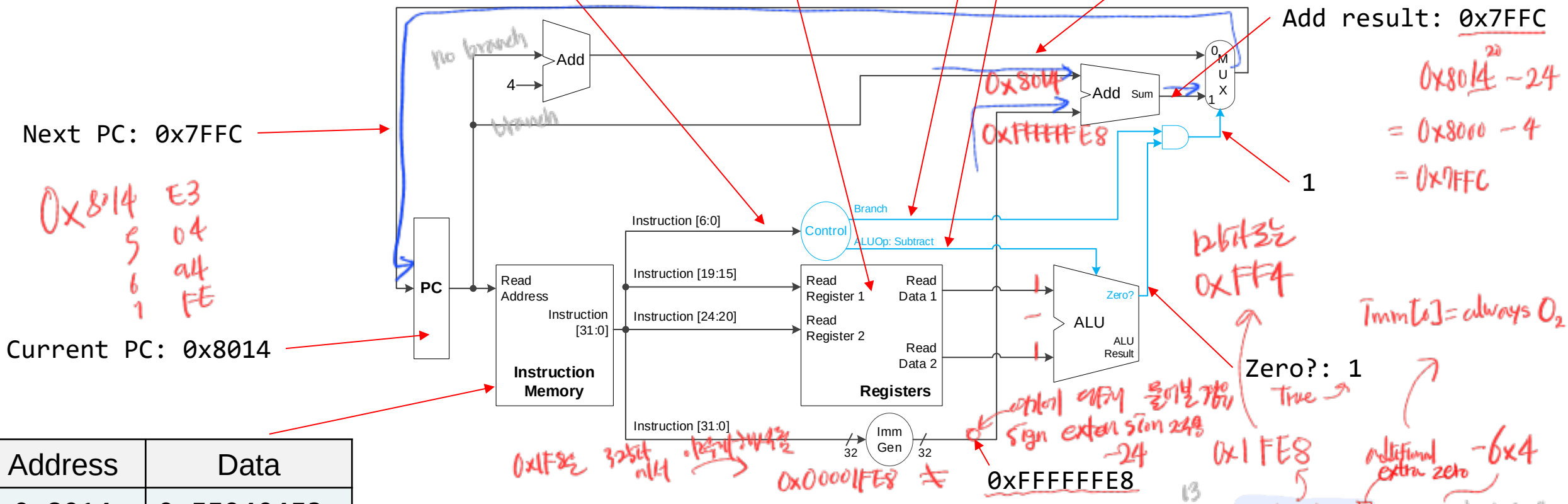
Opcode:1100011_2

PC+4: 0x8018

Add result: 0x7FFC

Next PC: 0x7FFC

Current PC: 0x8014



Address	Data
0x8014	0xFE9404E3

1111 1110 1001 0100 0000 0100 1110 0011

1 111111 01001 01000 000 0100 1 1100011

Imm[12] Imm[10:5] rs2=9 rs1=8 funct3 Imm[4:1] Imm[11] Opcode = beq

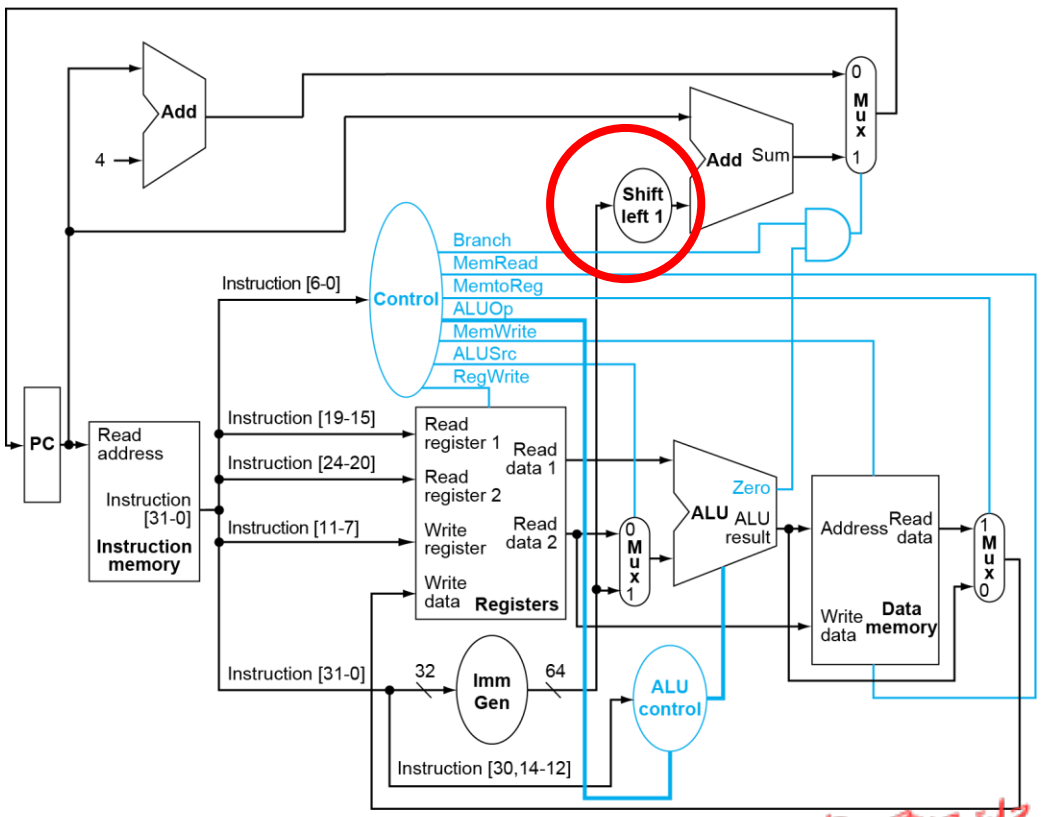
Imm[12:0] = 1111111101000_2 = -24_{10}

→ beq x8, x9, *LABEL*

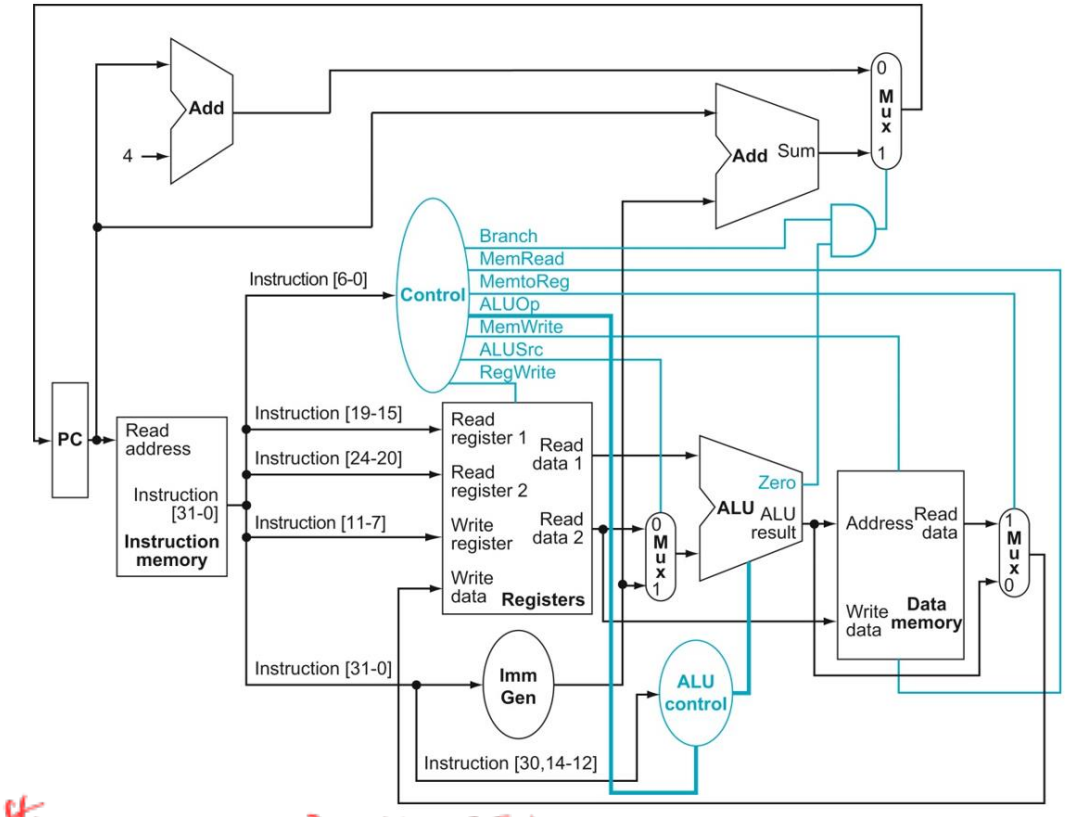
LABEL: points to 6 instrs before beq

TextBook 1st & 2nd edition Difference

- The CPU datapath drawn in the 1st edition has an additional “shift left 1” after Imm Gen



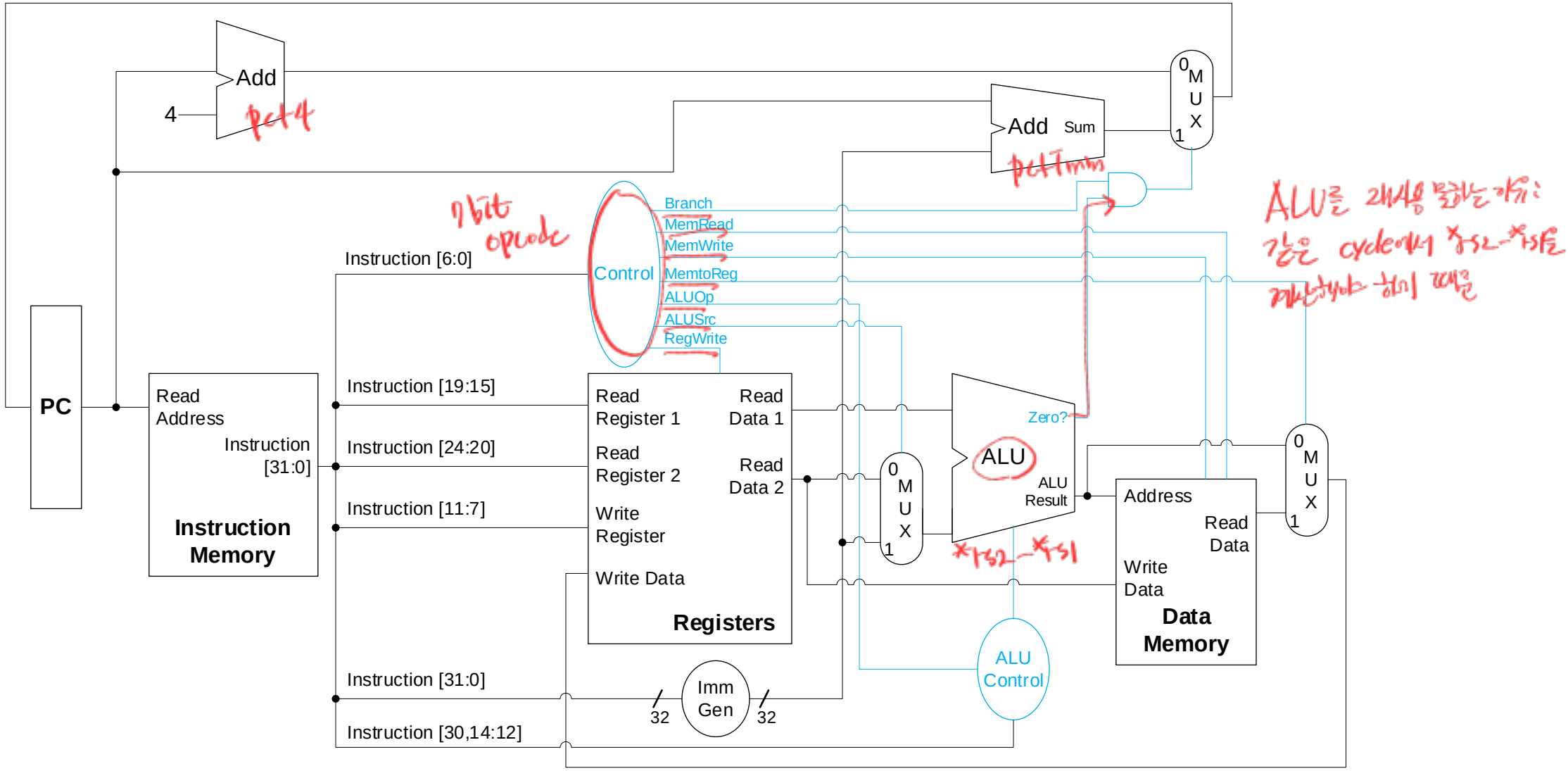
1st edition



2nd edition

- There is no need for a shift logic after the immediate value is fully assembled and sign-extended to 32-bit data.

CPU for R-type/Load/Store/Branch



Control Signals

Instruction	ALUSrc	MemtoReg	RegWrite	MemRead	MemWrite	Branch	ALUOp[1]	ALUOp[0]
Arithmetic (R-format)	0	0	1	0	0	0	1	0
lw	1	1	1	1	0	0	0	0
sw	1	X	0	0	1	0	0	0
beq	0	X	0	0	0	1	0	1

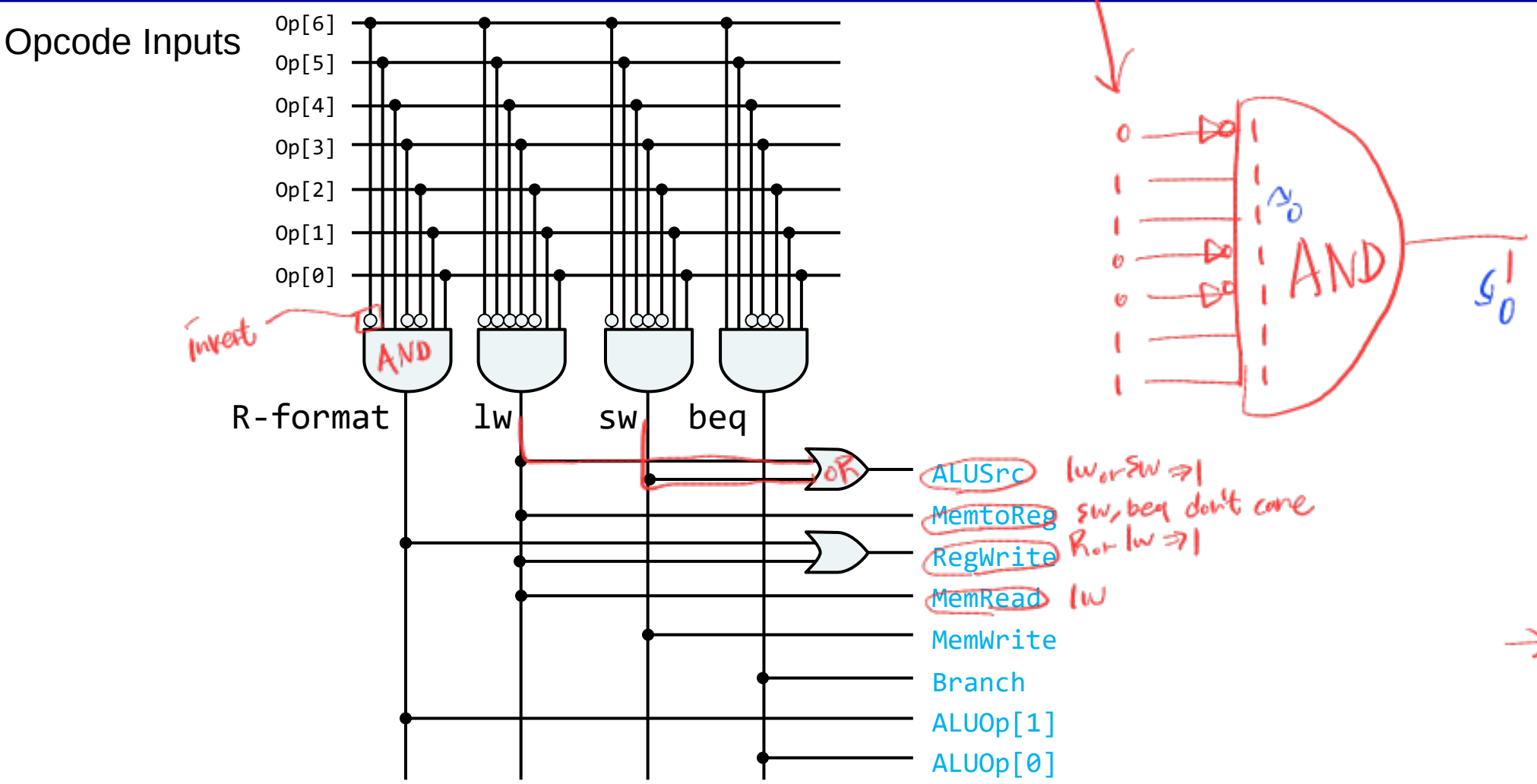
don't care

- Opcode
  - Arithmetic: 0110011
  - lw: 0000011
  - sw: 0100011
  - beq: 1100011

- Need to build a combinational logic!

Implementation of Combinational Control Unit

*Example not optimized

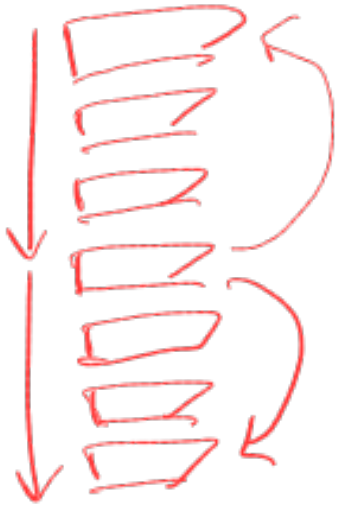


- Synthesis tool automatically optimizes RTL code → Combinational logic

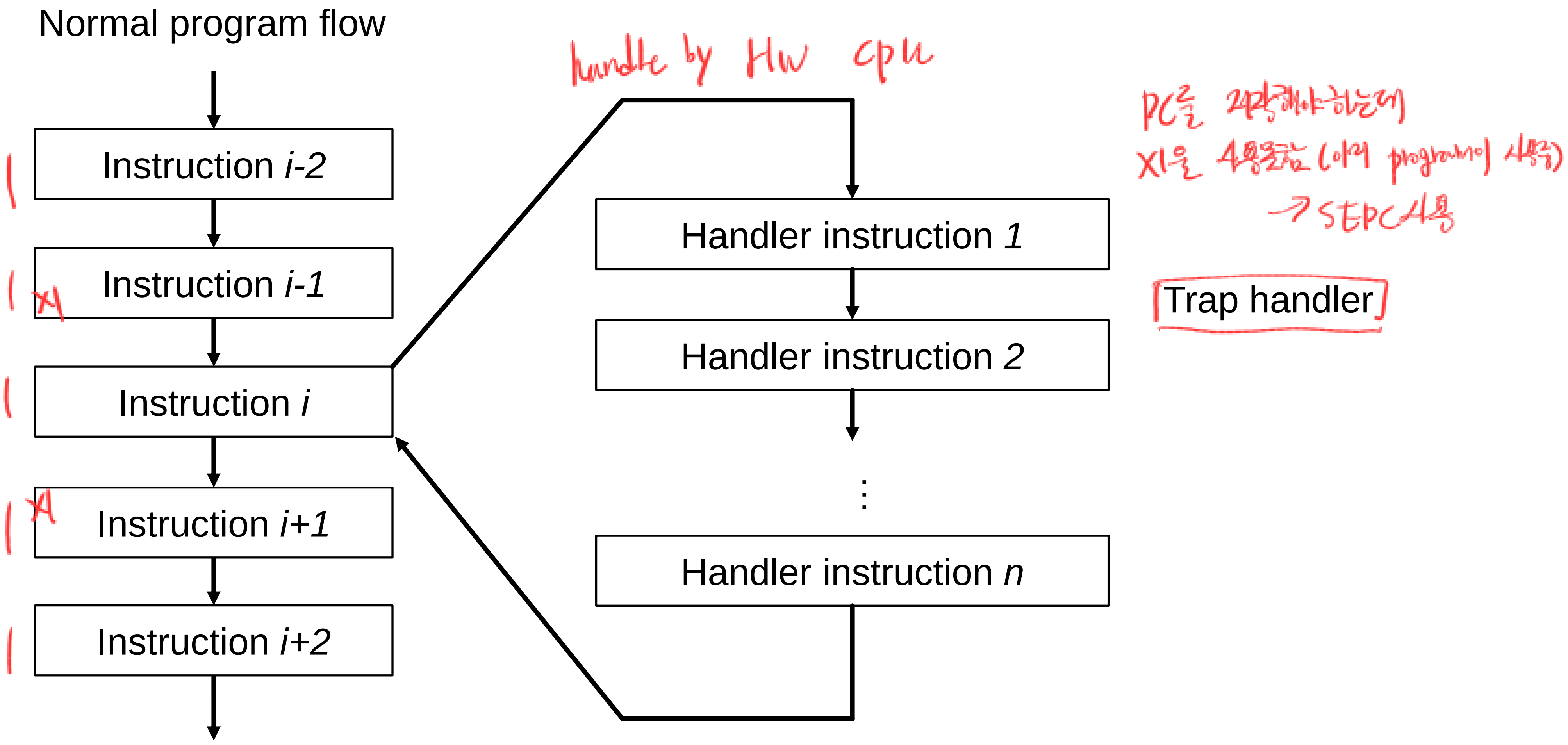
Exceptions and Interrupts

- “Unexpected” events requiring change in flow of control

- Exception
  - Arises within the CPU
    - e.g., undefined opcode, syscall, …
- Interrupt
  - From an external I/O controller



Altering the Normal Flow of Control



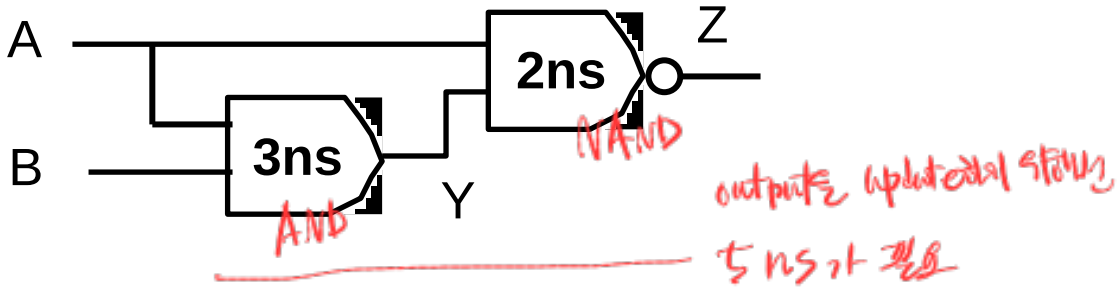
Handling Exceptions

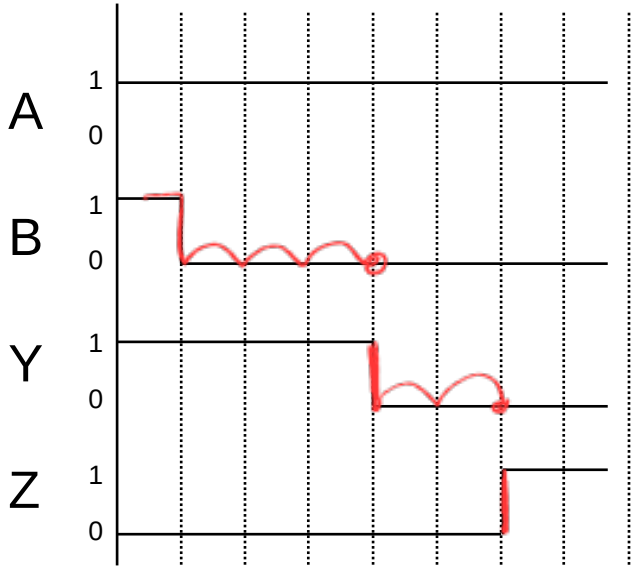
- Save PC of offending (or interrupted) instruction
  - In RISC-V: Supervisor Exception Program Counter (SEPC)
- Save indication of the problem
  - In RISC-V: Supervisor Exception Cause Register (SCAUSE)
- Jump to handler
  - Assume at 0x1C090000

Handler Actions

- Read cause, and transfer to relevant handler
- Determine action required
- If restartable:
  - Take corrective action
  - Use SEPC to return to program
- Otherwise:
  - Terminate program
  - Report error using SEPC, SCAUSE, …

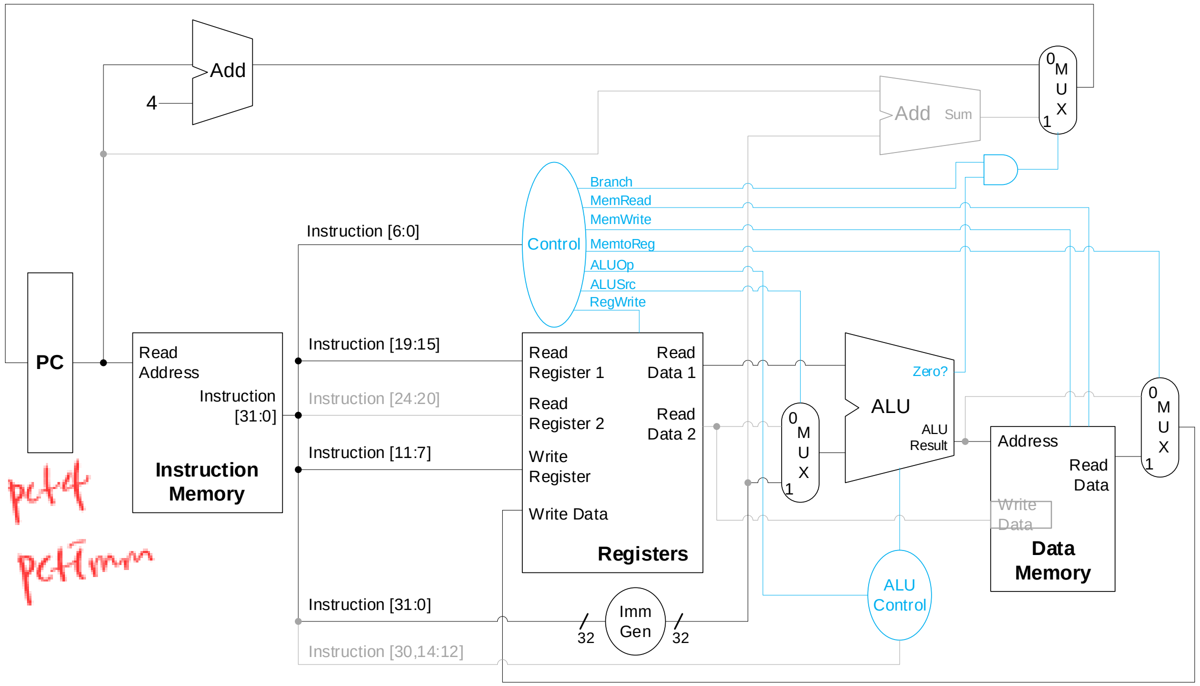
Combinational Logic Propagation Delay





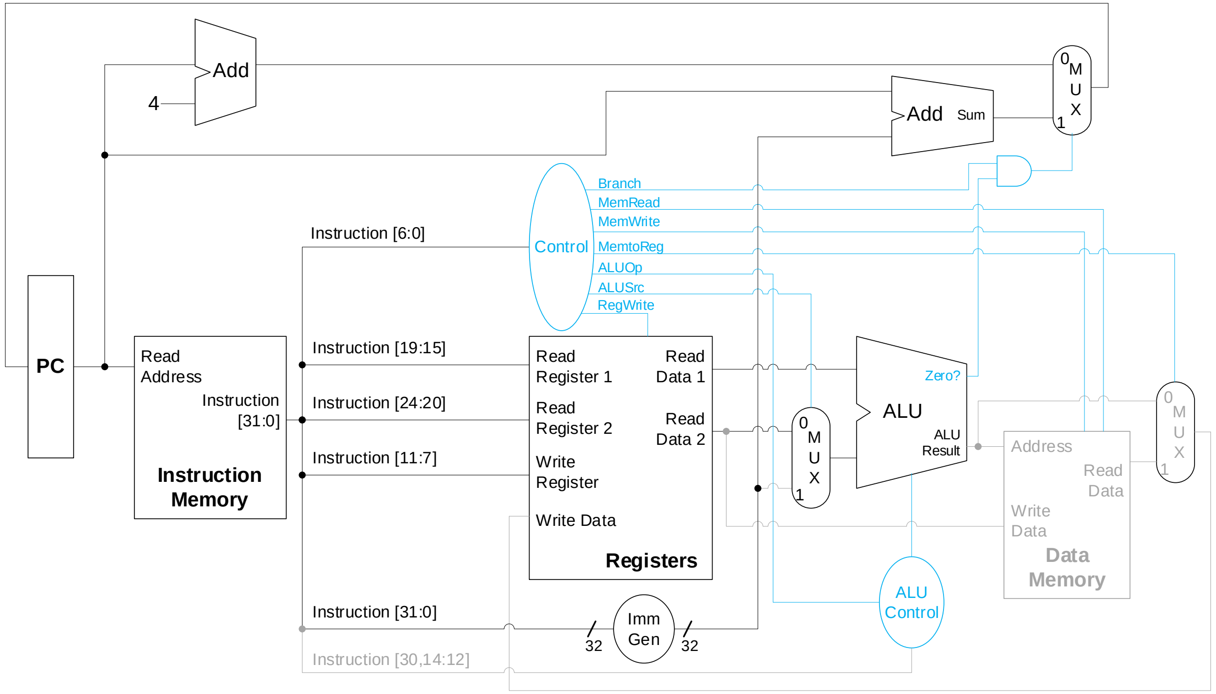
Propagation Delay

Single-Cycle for Every Instruction?



pc+4
pc+imm

LW Instruction



BEQ Instruction

800ps

500ps

300ps
낭비

instruction 따라 쓰는 단계가 다르고,
각각의 단계에 쓰이는 시간이 다름

clock period = 800ps → longest instruction에 맞춰짐

Pipeline Performance

- Assume the time for each stages is
  - 100ps for register read or write
  - 200ps for other stages

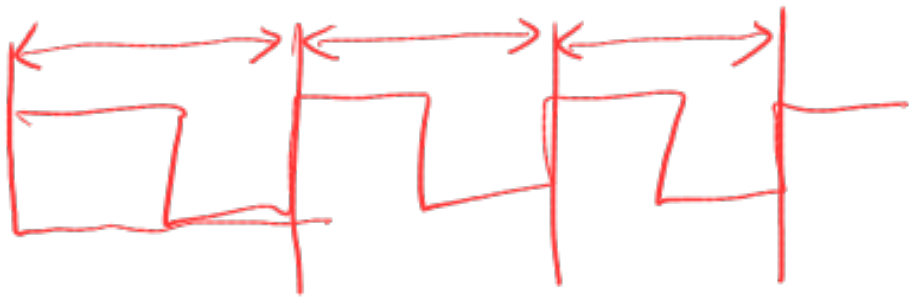
Instruction type	Instruction fetch	Register read	ALU op	Memory access	Register write	Total time
`lw`	200ps	100 ps	200ps	200ps	100 ps	800ps
`sw`	200ps	100 ps	200ps	200ps		700ps
`R-format`	200ps	100 ps	200ps		100 ps	600ps
`beq`	200ps	100 ps	200ps			500ps

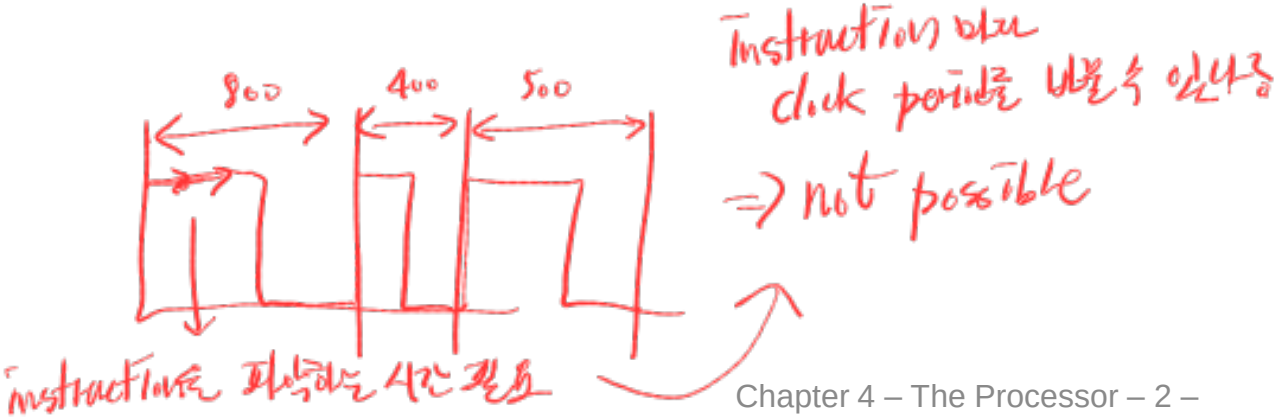
Performance Issues

- Longest delay determines the clock period
  - **Critical path**: The datapath that takes the **longest** time
  - Critical path in the single-cycle processor: load instruction
    - Instruction memory → register file → ALU → data memory → register file

- Not feasible to vary the clock period for different instructions

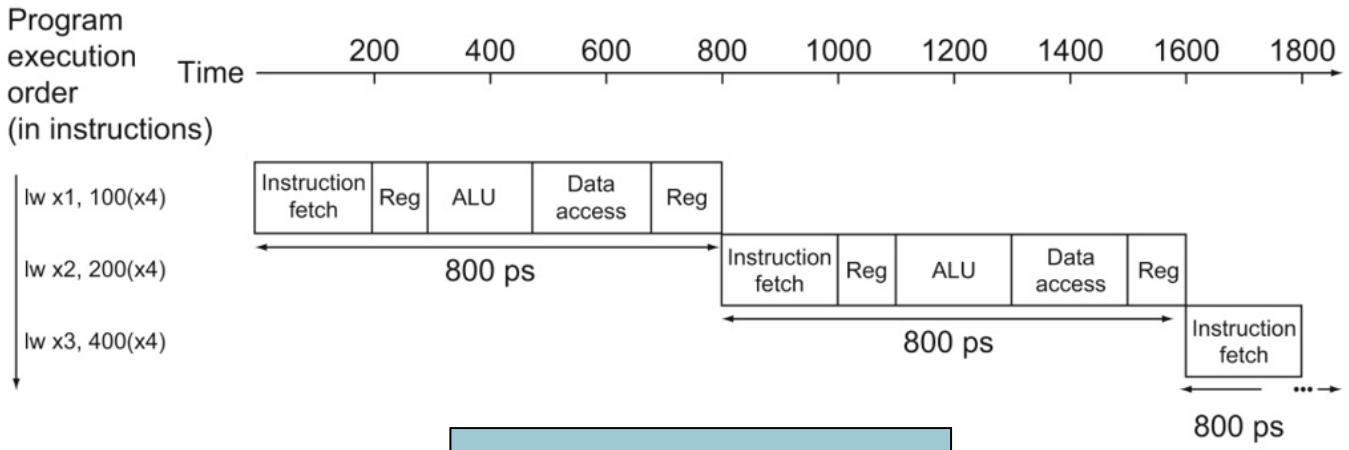
- We will improve performance using pipelining





Pipeline Performance

Single-cycle (T_c= 800ps)



Pipelined (T_c= 200ps)